

## UNIVERSITY PARK, PA, USA

WMO: 725128

Lat: 40.850N

Lon: 77.850W

Elev: 378

StdP: 96.87

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 54739

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -14.0      | -12.1 | -21.2                         | 0.6 | -12.0 | -18.4 | 0.8 | -10.8 | 11.9                  | 1.8  | 10.9 | -0.6 | 3.7                   | 270  | 0.473 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.7                   | 31.4            | 21.7 | 29.9 | 21.2 | 28.0 | 20.3 | 23.5                | 28.1 | 22.6 | 27.2 | 21.9 | 26.3 | 3.9                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 22.3                            | 17.8 | 25.3 | 21.3 | 16.7 | 24.5 | 20.8 | 16.2 | 24.2 | 71.9          | 28.1 | 68.5 | 27.5 | 65.8 | 26.8 | 26.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 10.2          | 8.5  | 7.5 | -16.4                      | 33.1 | 2.0                | 1.7 | -17.9                                              | 34.4 | -19.1      | 35.4 | -20.2      | 36.4 | -21.7      | 37.6 |    |    |
| (5)               |      |     | -17.2                      | 24.7 | 2.0                | 1.0 | -18.7                                              | 25.4 | -19.8      | 25.9 | -21.0      | 26.5 | -22.4      | 27.2 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 10.2   | -2.5  | -1.3  | 3.4   | 10.1  | 15.1  | 19.8  | 21.9  | 21.6  | 17.4  | 11.1  | 5.8   | -0.5  |  |  |
|                                                                        | DBStd       | 9.79   | 6.06  | 5.06  | 5.32  | 5.03  | 4.25  | 3.45  | 2.85  | 2.87  | 3.51  | 4.40  | 4.88  | 4.94  |  |  |
|                                                                        | HDD10.0     | 1498   | 388   | 318   | 216   | 58    | 6     | 0     | 0     | 0     | 1     | 39    | 143   | 328   |  |  |
|                                                                        | HDD18.3     | 3304   | 644   | 551   | 463   | 250   | 118   | 24    | 3     | 6     | 57    | 227   | 377   | 583   |  |  |
|                                                                        | CDD10.0     | 1582   | 2     | 1     | 11    | 63    | 166   | 294   | 369   | 359   | 223   | 74    | 17    | 3     |  |  |
|                                                                        | CDD18.3     | 347    | 0     | 0     | 1     | 4     | 19    | 69    | 114   | 106   | 30    | 3     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 2686   | 0     | 0     | 4     | 68    | 178   | 528   | 907   | 756   | 217   | 25    | 1     | 0     |  |  |
|                                                                        | CDH26.7     | 738    | 0     | 0     | 1     | 11    | 37    | 136   | 278   | 223   | 48    | 4     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 2.8    | 3.6   | 3.6   | 3.5   | 3.3   | 2.6   | 2.3   | 2.0   | 1.8   | 2.1   | 2.6   | 3.0   | 3.5   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 976    | 69    | 63    | 82    | 77    | 88    | 100   | 90    | 88    | 92    | 72    | 84    | 69    |  |  |
|                                                                        | PrecMax     | 1506   | 193   | 146   | 173   | 216   | 164   | 326   | 185   | 231   | 280   | 190   | 189   | 151   |  |  |
|                                                                        | PrecMin     | 712    | 12    | 9     | 24    | 9     | 22    | 8     | 20    | 10    | 22    | 2     | 17    | 14    |  |  |
|                                                                        | PrecStd     | 189    | 40    | 31    | 33    | 42    | 38    | 56    | 39    | 47    | 54    | 45    | 40    | 33    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 16.1  | 15.1  | 23.7  | 27.9  | 30.2  | 32.0  | 33.4  | 32.9  | 30.6  | 26.1  | 21.3  | 17.5  |  |  |
|                                                                        |             | MCWB   | 11.8  | 10.6  | 14.4  | 16.5  | 20.3  | 21.6  | 22.6  | 21.9  | 20.0  | 19.0  | 15.0  | 12.6  |  |  |
|                                                                        | 2%          | DB     | 12.8  | 11.1  | 18.6  | 25.2  | 27.7  | 30.0  | 32.0  | 31.2  | 27.7  | 22.8  | 17.8  | 12.7  |  |  |
|                                                                        |             | MCWB   | 10.3  | 6.3   | 11.4  | 14.8  | 18.5  | 21.3  | 22.3  | 21.8  | 19.1  | 15.4  | 13.2  | 8.9   |  |  |
|                                                                        | 5%          | DB     | 9.0   | 7.9   | 15.1  | 22.2  | 25.9  | 28.0  | 29.8  | 28.9  | 26.0  | 21.1  | 16.1  | 8.9   |  |  |
|                                                                        |             | MCWB   | 6.9   | 4.3   | 9.3   | 13.6  | 17.5  | 20.5  | 21.4  | 21.5  | 18.6  | 14.7  | 11.6  | 6.3   |  |  |
|                                                                        | 10%         | DB     | 5.5   | 6.1   | 12.3  | 18.9  | 22.9  | 27.0  | 27.9  | 27.4  | 23.8  | 18.7  | 13.8  | 6.2   |  |  |
|                                                                        |             | MCWB   | 3.4   | 3.2   | 7.8   | 11.6  | 15.9  | 19.8  | 20.5  | 20.2  | 17.8  | 14.5  | 10.2  | 3.7   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 13.7  | 11.2  | 15.1  | 18.5  | 21.5  | 23.9  | 24.9  | 24.6  | 22.3  | 19.9  | 16.8  | 14.0  |  |  |
|                                                                        |             | MCDB   | 15.3  | 14.6  | 21.4  | 25.6  | 28.3  | 28.3  | 30.3  | 28.2  | 26.7  | 25.2  | 19.4  | 16.4  |  |  |
|                                                                        | 2%          | WB     | 10.7  | 7.3   | 12.8  | 16.0  | 20.1  | 22.7  | 23.6  | 23.2  | 21.1  | 17.8  | 14.2  | 9.7   |  |  |
|                                                                        |             | MCDB   | 12.5  | 9.5   | 17.2  | 22.9  | 25.7  | 27.5  | 28.5  | 27.7  | 24.7  | 20.4  | 17.6  | 12.3  |  |  |
|                                                                        | 5%          | WB     | 6.7   | 5.0   | 10.3  | 14.6  | 18.9  | 21.8  | 22.7  | 22.4  | 20.1  | 16.1  | 12.4  | 6.7   |  |  |
|                                                                        |             | MCDB   | 8.8   | 7.8   | 14.9  | 20.1  | 23.6  | 26.4  | 27.3  | 26.7  | 23.3  | 19.9  | 15.3  | 8.3   |  |  |
|                                                                        | 10%         | WB     | 3.6   | 3.2   | 7.9   | 12.8  | 17.7  | 20.8  | 21.9  | 21.6  | 19.2  | 14.4  | 10.6  | 4.0   |  |  |
|                                                                        |             | MCDB   | 5.4   | 6.0   | 11.3  | 18.0  | 21.3  | 24.8  | 26.3  | 25.6  | 22.5  | 17.8  | 13.2  | 6.1   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 6.9   | 7.8   | 9.1   | 11.0  | 10.9  | 10.3  | 10.7  | 10.1  | 10.3  | 9.5   | 8.0   | 6.5   |  |  |
|                                                                        |             | MCDBR  | 10.4  | 11.2  | 13.3  | 15.8  | 13.8  | 12.4  | 12.5  | 12.4  | 13.4  | 13.6  | 11.3  | 10.2  |  |  |
|                                                                        | 5% WB       | MCWBR  | 8.7   | 8.6   | 8.2   | 8.3   | 6.6   | 5.5   | 4.8   | 5.0   | 6.3   | 7.4   | 7.9   | 8.5   |  |  |
|                                                                        |             | MCWBR  | 9.6   | 10.2  | 12.0  | 13.5  | 11.5  | 10.3  | 10.1  | 10.0  | 9.6   | 11.8  | 9.9   | 9.2   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.312 | 0.326 | 0.353 | 0.377 | 0.414 | 0.472 | 0.483 | 0.475 | 0.388 | 0.370 | 0.326 | 0.305 |  |  |
|                                                                        | taud        |        | 2.397 | 2.333 | 2.352 | 2.327 | 2.246 | 2.107 | 2.092 | 2.099 | 2.331 | 2.373 | 2.476 | 2.466 |  |  |
|                                                                        | Ebn at Noon |        | 841   | 881   | 896   | 894   | 865   | 812   | 798   | 792   | 848   | 820   | 821   | 823   |  |  |
|                                                                        | Edh at Noon |        | 82    | 101   | 112   | 123   | 136   | 157   | 158   | 153   | 113   | 97    | 76    | 71    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.63  | 2.50  | 3.45  | 4.54  | 5.13  | 5.63  | 5.70  | 5.06  | 4.12  | 2.81  | 1.92  | 1.33  |  |  |
|                                                                        | RadStd      |        | 0.12  | 0.30  | 0.30  | 0.45  | 0.54  | 0.48  | 0.38  | 0.29  | 0.52  | 0.32  | 0.22  | 0.17  |  |  |

## Historical Trends

|                              |       | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) Station Only            | DBAvg | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (47) Regional (33 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page